Instructions

The exercise must be developed individually and submitted via Blackboard by **02/10/2026**. The result should be a PDF report containing your answers and supporting images.

Objectives

This assignment aims to deepen the understanding of security properties, requirements, and threat models in real-world systems. By doing so, it enables the evaluation of the suitability and effectiveness of the various cryptographic systems discussed throughout the semester when applied to different practical scenarios.

Question 1

- Select one software, operating system, or service that you use frequently.
- Search for recent CVEs (last 12 months) related to the chosen software.
- Record the following information for three vulnerabilities:
  - CVE ID
  - Description
  - CVSS score and its interpretation
  - Associated CWE category
  - Exploit available (Yes/No)
  - Possible mitigations or fixes suggested
- Select one of the vulnerabilities found to deepen your analysis:
  - What is the impact of the vulnerability on a production environment?
  - Which systems could be compromised?
  - What preventive measures can reduce the risk?
  - Is there a patch available, and how can it be applied?
  - If there is no patch, suggest alternatives to mitigate the risk.

Question 2

Consider a credit card payment system in which three entities are involved in online transactions: the client, the credit card operator, and the seller. Describe the security properties that must be protected and the main threat classes to which the system is exposed.

Question 3

Cryptography is fundamental to data security. While modern cryptosystems are generally regarded as secure, related vulnerabilities continue to emerge. The following CVEs illustrate such issues. Based on these examples, analyze the main sources of vulnerabilities in cryptography-protected systems. Can you identify common CWEs related to these sources?

- CVE-2025-30406
- CVE-2025-52374
- CVE-2025-54887
- CVE-2025-3501

Question 4

Consider a remote employee connecting to a corporate network through a Virtual Private Network (VPN) using a protocol of your choice (e.g., IPsec, WireGuard, or OpenVPN). Describe the security properties that the VPN must guarantee, the cryptographic mechanisms it relies on for key exchange, authentication, and data confidentiality/integrity, and the main threat classes to which such a connection is exposed.

Question 5

Select a widely used network security protocol implementation (e.g., OpenSSL/TLS, OpenSSH, or an IPsec stack). Search for recent CVEs (last 12 months) affecting its cryptographic components, and record for three vulnerabilities:

- CVE ID
- Description
- CVSS score and its interpretation
- Associated CWE category
- Exploit available (Yes/No)

- Possible mitigations or fixes suggested

Select one of the vulnerabilities found to deepen your analysis:

- How does the vulnerability affect the confidentiality, integrity, or authenticity guarantees of the protocol?
- Which network scenarios or deployments are most at risk?
- Is there a patch available, and how can it be applied?
- If there is no patch, suggest alternatives to mitigate the risk while maintaining secure communication.